**Prakhar Gandhi — (+91)-9133234325 | gprakhar0@gmail.com | ResearchGate | Certified(micro1)**

Github | Portfolio | Website(Vikhorix Brand) | Utube | NanoTech | Multimodal ML Pipeline | Automation | ChatBot | MapReduce(Universal TopK/BottomK) | Scalable Image Features App | Awesomest Webapp | Shor's Algorithm | AGI | Quantum ML | Universal Algorithms Visualizer | Scalable Distance Metrics | Scalable Folder Search | Scalable OCR | Software Dev | Agentic AI | Automate Imputation | Universal HTML Code Generation | Material UI | World Travel Route Explainer | Optimized Universal Dashboard Automation | Universal ML Explainer | Universal EDA | Optimized Computer Vision Utilities | Retry Mechanism Automation | Chaining fn Automation | Flowchart Automation | Scalable Pagination API | Universal Graph Analysis | TextGrouper | Run Universal Functions | Kaggle Level Accuracy | Automate Steps | Universal GET/POST FastAPI | Scalable Docs Generation | Scalable Pretty Print

## Summary

Lead AI Engineer with 6+ years of experience in Python, ML, and Generative AI. Proven record of building scalable ML pipelines, agentic AI systems, and internal products used by 1000+ users. Strong background in NLP, computer vision, and parallel computing with measurable performance gains. Experienced in mentoring and cross-team leadership.

## Education

**BITS Pilani - B.E. in Electronics and Instrumentation** Hyderabad, Telangana
*July 2015 – Dec 2019*

## Academic & Professional Development

**Coursework**: Data Structures and Algorithms, Neural Networks and Fuzzy Logic, Discrete Math for Computer Science
**Self Taught Courses**: Database Management Systems, Number Theory: Project Euler
**Certifications**: Deep Learning: Advanced NLP and CNNs, RNNs, LSTMs and GRUs, Bayesian Machine Learning in Python: A/B Testing, Recommender Systems and Deep Learning in Python, CNN in python - Computer Vision
**Certificates Link**
**Awards**: Have received a gold award from Standard Chartered Bank Global Business Services Private Limited as a part of AI/ML Hackathon, Victory League Award in Ericsson

## Technical Skills

**Programming Languages**: Python, C++, JavaScript
**Machine Learning & AI**: NLP, Computer Vision, Generative AI, RAG, Model Evaluation, Feature Engineering
**Frameworks & Libraries**: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, LangChain, Langgraph, Crew AI
**Backend & APIs**: Dash, Django, Fastapi, Microservices, Tool calling functions for CRUD
**Data & vector databases**: Qdrant, Snowflake, Testing(Playwright) & Documentation Automation(pdoc)
**Cloud & DevOps**: Docker, Terraform, Azure Functions, Agentic AI using Azure Foundry, CI/CD, Amazon Q, Github Copilot.

## Experience

**Technical Lead Architect (AIML Computational Science Specialist - Gen AI)** April 2026 – Present
*Accenture - • AT&T • Hunter Douglass Solution • Campbell • Linde Voyager • Coty • Diebold* *Bangalore, IN*

- Scaled up the differentiation between scanned and digital pdfs.
- Scaled up the contract handling using guardrail mechanism/custom field validation using fuzzylogic+easyocr.
- Designed a production-grade CRUD pipeline integrated with Snowflake for contract extraction, reducing manual processing effort.
- Used RAG to summarize text and scaled it upto million text datasets.
- Explored skills in Agentic AI like claude and Advanced Agentic AI.(Planner Agents, Graph construction, Orchestration, MCP)

**Performance Metrics – Key Achievements**
*Accenture* *Bangalore, IN*

- Project Delivery & Technical Contribution
  - Contributed to the **AT&T Agentic Contract Implementation** initiative, working on automation, document processing, extraction, validation, and retrieval-related activities.
  - Completed multiple sprint deliverables, including:
    - **Rerouting logic scalability** – progressed to 80% during Sprint 1.

        - **PDF duplicate detection using hashing** – completed 100%.
    - Implemented **multi-contract-type document handling**, including automation and table extraction, followed by scalability enhancements.
    - Developed functionality to recursively process PDF files from folders and validated the implementation against **EPA contract datasets**.
    - Completed Sprint 3 activity involving **LLM response validation**, including testing against raw ETF contract text and sharing findings with the team.
- Contract Analysis & Data Validation
    - Performed analysis of **scanned/OCR contract documents**, including completion of analysis for **100 documents through team-level work distribution**.
    - Created **BICI analysis documentation and variation lists** and shared the findings with the team.
    - Worked extensively on **AVPN contract validation**, comparing LLM-generated JSON responses against actual contracts.
    - Analysed approximately **40 documents initially**, followed by additional sets of **200–250 documents**, while identifying issues and modifying prompts based on findings.
    - Worked on identifying important contract sections such as **Credit, Rates and Discounts**, and used the findings for prompt-generation activities.
- LLM Validation & Guardrails
    - Implemented and worked on multiple validation mechanisms for LLM-generated responses:
        - Strict **JSON Schema validation**.
        - Data constraints such as regex, enums, and numeric bounds.
        - **Cross-field validation** for contractual consistency.
        - Automated retry/self-correction mechanism using validation errors.
        - Error-logging structure for tracking validation failures by contract type.
    - Contributed toward establishing a more reliable and controlled **LLM-based contract extraction and validation process**.
- Automation & Scalability
    - Focused on converting manual/document-processing activities into **reusable and scalable automation**.
    - Enhanced PDF processing through recursive folder handling and tested the solution against contract datasets.
    - Worked on scaling extraction solutions for important sections of PDFs and on implementing guardrails for extraction and retrieval.
    - Contributed to automation of multi-contract-type document processing and table extraction.

**Senior AI Engineer (Python Developer - Generative AI)** Feb 2025 – April 2026

*Wipro - Ericsson India Private Limited* *Bangalore, IN*

- Developed a ChatGPT-style user interface with CRUD features in Dash + SQLite
- Demo presented to 1000+ Ericsson employees.
- Quality Features included Pagination, Dark/Light Theme Toggle, Export to CSV support for tables, Interactive Zoomable/Drillable Graphs (Plotly), Textual/Tabular/Graph formatting.
- Implemented Agentic AI workflows using MCP, LLM-Based Evaluation, RAG, pydantic, chunking and retrieval, Reinforcement Learning Human Feedback, recursive text splitter, reranker mechanism, langchain, openai-models(GPT-4/3.5, Groq), Planner Agents, Graph construction, Orchestration Agent.

**Senior Software Engineer (Python Developer)** May 2024 – Jan 2025

*Wipro - Mercedes-Benz Research and Development India* *Bangalore, IN*

- Automated manual cpp file code generation by doing Parameter and function name extraction of car product cpp files using python and multiprocessing. Optimized 2nd level folder search under 2 mins using parallel processing.
- Developed a Flask application that includes live terminal using socketio and docker integration in end to end GUI.

**Data Analyst 2** Aug 2020 – Jan 2024

*Standard Chartered Bank Global Business Services Private Limited India* *Bangalore, IN*

- Automated Backward Elimination ML procedure for correlated features removal.
- Developed a Tensorspace visualization custom component to solve the memory leakage issues at hackathon.
- Developed a flask app to visualize model predictions and flipped classes based on synonym replacement, insertion and deletion of tense words of banking words present in a general spacy dictionary to check the robustness of model to any such attacks.

- Implemented multiprocessing on django microservices by using parallel processing using pandarallel package and decreased the execution time.
- Developed a QA generation pipeline from scratch using the knowledge of competitive programming by invoking the Large Language Models named t5-base-squad of medium parameter size. Speedup the entire process by using parallel processing.
- Automated synthetic dataset generation by doing Multi blending labels of product documents(watermarks, barcodes, seals and other confidential details). This work involved Parallel Processing improved Data Creation for the agreed deliverables from 4 hour to 1 hour per document.
- Implemented McNemar Test via statistical analysis for cross-team members from structured data to compare the performance of 2 classifiers and selecting the best model further integrating it within the champion challenger pipeline.

## Projects - Standard Chartered

**NER Explainability Creation** | *Python, sklearn, pandas, numpy, plotly, dash* SCB

- Built local and global NER explainability visualizations including entity probability, POS importance, and state transition analysis.
- Implemented an end-to-end NLP pipeline for an ML-based NER explainer search engine.
- Deployed CRFsuite NER explainability using Dash, enabling accurate class-wise analysis at sentence level.
- Designed a scalable Champion–Challenger framework across NER, regression, classification, and object detection pipelines.
- Optimized evaluation metrics and system performance to support pipelines at million-scale data volume.
- Presented model explainability and performance insights to senior leadership.

**Image duplicate prediction** | *Python, pandas, numpy, tensorflow(imagededup), pytorch(detecto), opencv* SCB

- Built ML-based duplicate detection systems for banking PDFs, improving identification of barcodes, seals, and key document artifacts by 80%.
- Developed object detection models for barcodes, watermarks, and signatures using Faster R-CNN with SSIM-enhanced similarity scoring, achieving 85% accuracy.
- Generated analytical visualizations and client-ready reports to communicate model performance and validation results.

## Experience with Assignments (External)

**Zycus assignment** | *Python, pyspark, ML* Zycus

- Built a PySpark text classification pipeline for product reviews, achieving a 91% speedup **Github Link**

**Happymonk paper implementation** | *Python, pandas, numpy* Happymonk

- created a 3 layer artificial neural network from scratch as a part of research paper provided to me in coding assignment and gained 96% accuracy for benchmarks like iris/mnist dataset, **Github Link**

## Achievements

**Competitive Programming Ranks** | *Python, Cpp* Contests

- Dark Red Coder(Greatest color) on Atcoder in the history of competitive programming which no legendary grandmaster has achieved till date, name's monkeyDBoa. **Link**

**System Design, Cloud Architecture, Distributed Systems, AI Architecture, Leadership** At Companies

- Mentored a junior colleague in Standard Chartered with dataset preparation from pdfs using easy ocr.
- Mentored a fresher in Ericsson with automation of test cases using robot framework with playwright.
- Mentored a senior colleague in Standard Chartered as a part of hackathon in visualizing robustness check of ML/DL model at a targetted classwise image level attack and made a opacity test to compare a permutation of two classes at a time by implementing the same from the paper. Further Detected poison in image dataset.

**Hall of Famer!** | *Python*

- Picasso Themed Scalable Blockchain Webapp without use of database to keep the randomization of nft images intact with encryption algorithm to keep passwords secure, added buttons and scaled it up to billion fold world,
  **1) Github Link 2) Python Package if you prefer to run in local 3) Deployed Demo**
- Implemented Love Hate Algorithm with face detection to find best colleague to work with and scaled it up,
  **1) Github Link 2) Python Package if you prefer to run in local**